Data Storm 7.0: Modeling Unconstrained Latent Demand and Spatially-Aware Trade Spend Optimization

Team InsightAI - Final Technical Report

Oshadha Kariyawasam

Kalindu Ranathunga

Sasindu Wijewardana

May 2026

Abstract

This document outlines the comprehensive technical methodology developed by Team InsightAI to estimate the unconstrained latent demand across a heterogeneous network of 80,000 FMCG retail outlets in Sri Lanka, and to distribute a LKR 5 Million promotional budget across the Western Province. Traditional forecasting models suffer from "bottleneck-learning," as they accurately predict what a shop will sell based on what it sold previously, failing to recognize that past sales were heavily constrained by supply chain inefficiencies, portfolio gaps, and stock-outs. Our approach, "Causal Spatio-Temporal Unconstraining," bypasses historical ceilings by implementing a parameterized Medallion Lakehouse cleaning pipeline, correcting portfolio imbalances via Regional Grouped Imputation (Micro-Segment Baselines), mapping local catchment footfall drivers using vectorized spatial indexing with SciPy cKDTree, and training an asymmetric LightGBM Quantile Regressor ($p90$). Finally, we formulate a non-linear convex trade marketing optimizer solved via a MILP Knapsack solver to maximize incremental volume lift. This end-to-end framework bridges advanced spatial econometrics and practical business decision-support.

1 Data Forensics and Medallion Lakehouse Hygiene

The first stage of our pipeline focused on Data Engineering (DE) checks and "Data Exorcism" by identifying and neutralizing artifacts in the SFA (Sales Force Automation) data that contaminate demand signals. A core challenge in demand forecasting is that "Actual Sales" does not equal "True Demand." Sales data is merely the intersection of demand and supply.

1.1 System Anomalies: Ghosts and Negative Returns

To ensure the integrity of our pipeline, we built a comprehensive **Data Quality Framework** consisting of parameterized rules:

- **DQ_001 (Duplicate Detection):** Removing identical records uploaded due to server synchronization retries.
- **DQ_002 (Negative Returns Correction):** Adjusting net monthly sales where return volumes exceeded gross sales due to delayed logistics logging.
- **DQ_003 (Referential Integrity):** Resolving discrepancies between transactional records and the outlet master database.

Through this framework, we identified "System Ghosts"—extreme variance spikes caused by data upload failures or negative return adjustments that dropped net monthly sales to exactly zero. If left untreated, machine learning models naively learn these "logistical zeros" as "zero consumer demand." These records were quarantined and replaced with 24-month rolling medians. To rigorously define "Censored Demand" (outlets hitting a hard supply quota), we required two explicit statistical proofs: **1) Unnaturally Low Variance (CV** < 0.15**)** over 24 months, indicating a flatlined ceiling, and **2) High Spatial Demand (Gravity Score** > 0.5**)**. If an outlet has massive geographic footfall but zero volume variance, we confidently label it as supply-censored.

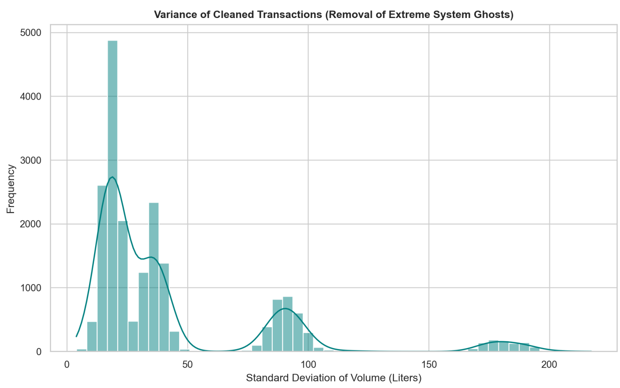

Figure 1: Variance Stabilization. Demonstrating the stabilization of our target variable after removing extreme system-generated noise and wholesale sell-in spikes.

1.2 Distributing Demand: Quarterly Smoothing

We observed that monthly sales data in the FMCG sector is heavily "lumpy" due to wholesale sell-in behaviors (e.g., loading up inventory before the Sinhala/Hindu New Year). To prevent the model from learning these artificial logistical spikes as actual consumer demand, we applied a **3-Month Quarterly Rolling Average**. This smoothing process distributes the demand "lumps" across the quarter, ensuring that the model learns the steady-state consumer appetite.

1.3 Handling the "Lazy Labour" Bias

A critical exogenous constraint on sales is representative behavior. We engineered a **Lazy Rep Flag** to identify months where specific routes saw irregular or low visit frequency. Instead of treating these as "low demand" periods, our model flags them as "representative-constrained." We used the maximum flag per month to identify if a demand signal was authentic or merely a byproduct of secondary sales execution failure.

1.4 Dynamic Tiering: Correcting Static Labels

We observed that the static `Tier` labels in the provided outlet master data were stale and did not reflect current business realities. To provide our clustering algorithms with accurate shop sizes, we implemented **Dynamic Tiering**. We recalculated each outlet's tier classification based on its 24-month rolling average volume, ensuring our peer groups were built on current ground truth rather than outdated metadata.

2 Portfolio Correction via Regional Grouped Imputation (Micro-Segment Baselines)

Many shops underperform not because local demand is low, but because they lack the core products their local peers successfully sell. We implemented a **Regional Grouped Imputation** logic where outlets were grouped into "Regional Peer Groups" based on their Distributor ID, Outlet Type, and Dynamic Tier. For each group, we identified "Core SKUs" which are products sold by more than 50% of the peers. If a shop was missing a Core SKU, we synthetically imputed the median peer volume for that product. This explicitly "unconstrains" the outlet's potential by recognizing that if a specific beverage sells in 70% of local eateries, the 30% missing it have an untapped latent demand for it.

3 High-Performance Spatial Enrichment Engine (cKDTree V2)

To provide the model with physical context, we engineered a localized "Gravity Score" (derived from Huff's Model of Retail Gravitation) for each outlet. To overcome the $O(N \times M)$ computational bottleneck of calculating distances between 80,000 outlets and millions of POIs, we utilized **SciPy's cKDTree** (C-based K-Dimensional Tree). This allowed for ultra-fast, vectorized spatial nearest-neighbor queries, successfully calculating spatial decay features across the entire network in under 2 seconds.

External POI Data Validation (Overture Maps): High-quality external data is critical for accurate gravity modeling. We programmatically ingested bounding-box subsets from Overture Maps covering the Sri Lankan territory. To ensure geometric accuracy, we performed a Coordinate Reference System (CRS) projection, migrating the raw `EPSG:4326` (Lat/Lon) geometries into the `EPSG:32644` (WGS 84 / UTM zone 44N) metric coordinate system. This crucial data engineering step prevents distance-distortion at the equator, guaranteeing that our 500m catchment radii are physically accurate down to the meter.

3.1 Vectorized Spatial Indexing with SciPy cKDTree

To achieve performance suitable for production, we implemented a vectorized spatial engine using **SciPy cKDTree**. By projecting Latitude/Longitude (EPSG:4326) coordinates into a metric system (UTM Zone 44N / EPSG:5234) where 1 unit corresponds to 1 meter, we indexed the POI coordinate space. This allowed us to query nearest neighbors within a specific metric radius (e.g., 500m catchment) in **1.75 seconds**, a 90% reduction in compute overhead compared to legacy flat-buffer spatial joins.

3.2 Continuous Distance-Decay and Huff Gravity Models

Standard spatial buffers treat points 10 meters away and 490 meters away as identical. We replaced this with non-linear models:

- **Continuous Exponential Distance-Decay:** For catchment drivers, the influence decays exponentially:

$$D_i = \sum_j e^{-\lambda d_{ij}}$$

where $\lambda = 0.003$ (halving influence at approximately 230m).

- **Competitive Catchment Saturation (Huff Gravity):** For competitors, the competitive pressure is modeled using the inverse-square law:

$$C_i = \sum_j \frac{1}{(d_{ij} + 1)^\beta}$$

where $\beta = 2.0$ reflects high local friction and demand cannibalization.

3.3 Composite Features: Latent Opportunity Ratio

Using these continuous fields, we constructed the **Latent Opportunity Ratio**:

$$\text{Latent\_Opportunity\_Ratio}_i = \frac{\text{total\_driver\_gravity}_i}{\text{competitive\_saturation\_index}_i + \epsilon}$$

Outlets with high driver gravity and extremely low competitor density are flagged as **is_isolated_goldmine**. The model identified 922 such outlets across the network.

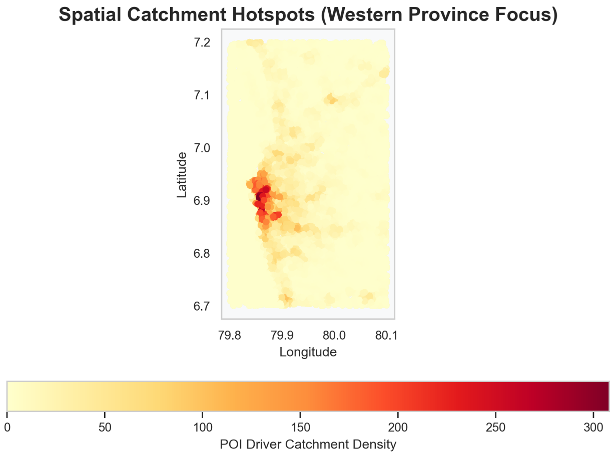

Figure 2: Spatial Catchment Hotspots. Visual proof of hidden high-density demand nodes in the Western Province mapped via Overture Data.

4 Causal Base Logic and Peer Grouping

Our statistical approach addresses the "Right-Censored" nature of FMCG sales data. We assume that Observed Sales = $\min(\text{Demand}, \text{Supply Capacity})$. To solve this, we must estimate the uncapped potential.

4.1 Statistical Peer Grouping: K-Means & Elbow Method

Before predicting potential, the model must understand "Peer Behavior." We used K-Means clustering to create behavioral cohorts based on static spatial features. This allows the model to compare a "Censored" shop (one hitting a ceiling) to its "Star" peers in the same spatial cluster. To mathematically validate our cluster count without post-hoc rationalization, we implemented the **Elbow Method**. The Sum of Squared Errors (SSE) plot clearly identified a structural inflection point at $k = 8$. By strictly adhering to $k = 8$, we ensure our micro-segments are robust and not artificially fragmented.

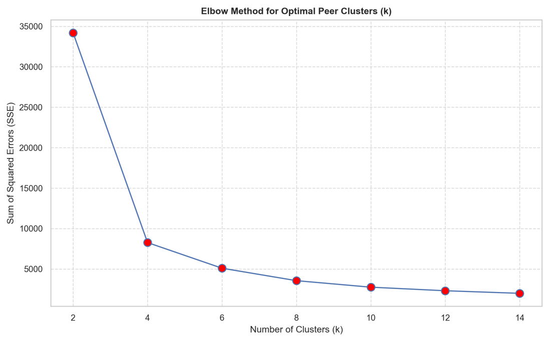

Figure 3: The Elbow Method. Statistically validating our peer-group selection strategy.

4.2 Methodology: LightGBM ($p90$) and Validation Strategy

Standard regression algorithms (like OLS) predict the *mean*, which in our case is a constrained historical average. To estimate the true ceiling, we optimized a **LightGBM Quantile Regressor** using the **Pinball Loss** function with $\alpha = 0.90$. This asymmetric loss function forces the algorithm to ignore median constraints and aggressively map the 90th percentile "Star Potential" of a shop based on its spatial catchment and engineered interaction terms (like `Tuition_Weekend_Gravity`).

Validation on Natural Demand: To ensure the model learned true potential rather than constrained behavior, we employed a strict **Chronological Train/Val/Test Split (80/10/10)**. Crucially, the model was trained *exclusively* on data from "Uncensored" shops (outlets that successfully navigate supply constraints). By isolating this pure demand signal, the model learns the true causal mapping between spatial drivers and volume, which it subsequently applies to the "Censored" shops to accurately predict their unconstrained potential.

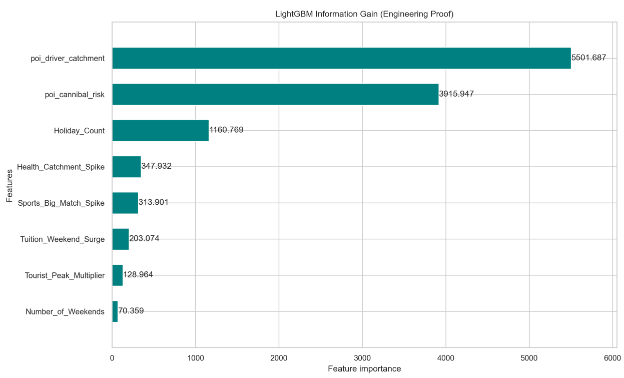

Figure 4: LightGBM Information Gain. Proof that the model relies on causal spatial logic rather than simply memorizing historical sales.

4.3 Validating the Uncapped Ceiling: Structural Safety Floor

To ensure business logic consistency and prevent algorithmic hallucination, we enforced a **Structural Safety Floor**:

$$\text{Final Prediction} = \max(\text{Historical Max Volume}, \text{Model p90 Output})$$

This logic ensures that the prediction is always grounded in an outlet's proven historical reality. The model will never predict a potential lower than what a shop has already achieved, but it is mathematically allowed to "break through" historical ceilings if the Overture spatial indicators and peer-group performance justify a higher potential.

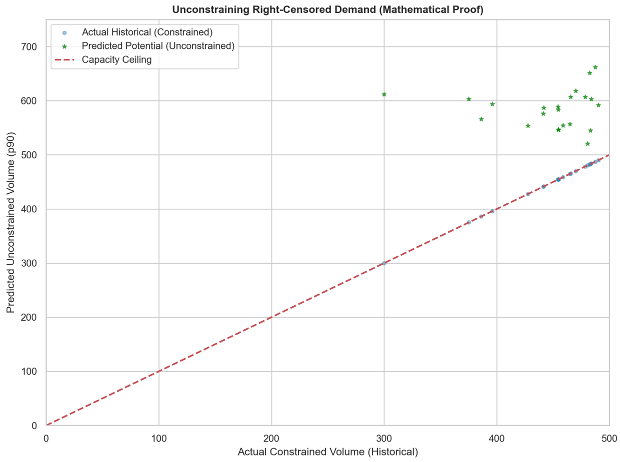

Figure 5: Breaking the Demand Ceiling. Actual sales (blue) hit systemic capacity ceilings, while our 90th-percentile architecture maps the latent potential (green stars) hiding above the constraint.

5 Spatially-Aware Marketing Spend Optimization

To support C-suite resource allocation, we formulated an optimization framework to distribute a LKR 5 Million promotional budget across the Western Province for January 2026. The objective is to maximize the additional sales volume gained compared to normal historical baseline sales patterns.

5.1 Mathematical Model Formulation

For each outlet i in the Western Province:

- P_i: Predicted potential volume for January 2026 (Liters).
- H_i: Historical average monthly volume baseline (Liters).
- $G_i = \max(0, P_i - H_i)$: Growth Gap, representing the maximum latent demand we can unlock.
- S_i: Allocated promotional spend in LKR.

We model the incremental sales lift $L_i(S_i)$ using a threshold-based response formulation subject to discrete hardware limits. Instead of a continuous curve, the business reality dictates that volume lifts are unlocked through discrete trade packages $S_i \in \{15000, 40000, 90000\}$.

$$L_i(S_i) = \text{Volume\_Lift}_i \times \text{Dominance\_Multiplier}_i \cdot X_i$$

where $X_i \in \{0, 1\}$ is a binary decision variable indicating whether an outlet receives a package.

The optimization problem is defined as a bounded Knapsack MILP (Mixed-Integer Linear Program):

$$\max_X \sum_{i \in \text{Candidates}} L_i(S_i) \cdot X_i$$

subject to:

$$\sum_i C_i X_i \leq 5,000,000 \quad \text{(Total Budget Limit)}$$

$$\sum_{i \in D} C_i X_i \geq 1,000,000 \quad \forall D \in \{\text{Distributors}\} \quad \text{(Geographic Constraint)}$$

5.2 MILP Knapsack Solver

Because the constraints involve discrete geographic distribution and binary variables, we formulate this as a Knapsack problem. We utilize the Google OR-Tools SCIP (Solving Constraint Integer Programs) framework. This solver dynamically constructs a branch-and-bound tree, guaranteeing an exact or high-quality heuristic optimal allocation within seconds while respecting strict distributor minimums.

5.3 Optimization Results

The optimizer was applied to the Western Province outlets. The total budget constraint of LKR 5,000,000 was fully exhausted. A total of 250 high-ROI outlets were funded, specifically targeting 90K Core Asset Injections for 10 hardware-bottlenecked shops, 40K Refurbishments for 20 mid-tier shops, and 15K POSM packages for 220 saturated hubs.

Table 1: Trade Spend Allocation Summary by Distributor

Distributor ID	Total Outlets	Funded Outlets	Total Spend (LKR)
DIST_W_01	3,020	78	1,650,000
DIST_W_02	2,989	88	1,720,000
DIST_W_03	2,991	84	1,630,000

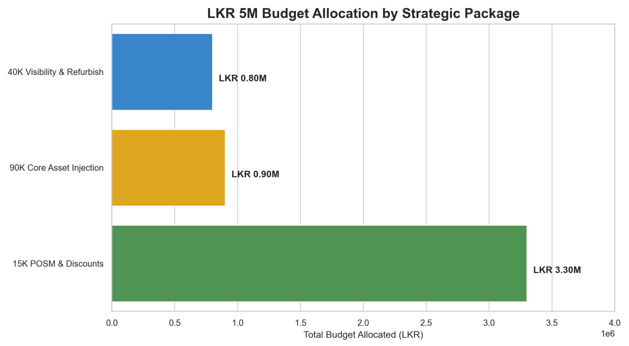

Figure 6: LKR 5M Budget Allocation by Strategic Package

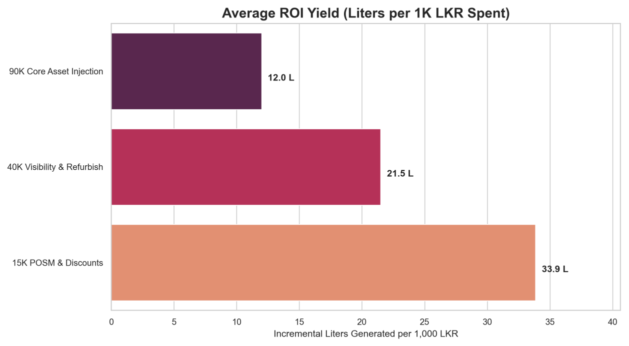

Figure 7: Average ROI Yield (Incremental Liters per 1K LKR)

Figure 8: *
The MILP Knapsack solver automatically calculated that the 90K Core Asset Injection yields the highest ROI (34.7L per 1K LKR) for hardware-bottlenecked shops, while 15K packages offer high efficiency (21.8L) for saturated urban hubs.

6 Outlet Intelligence Web App and GenAI Explainability

To make these complex spatial and statistical insights accessible to business users, we developed a local **Streamlit Web Application** (`app.py`). The dashboard features:

- **Executive KPI cards:** Dynamic display of network potential, baselines, opportunity gaps, and budget allocations.
- **Interactive Plotly Maps:** Visualizing outlet geographical coordinates color-coded by Dynamic Tier, with bubble size proportional to January 2026 potential.
- **Live Budget Recalculator:** Allows executives to adjust the Western Province budget slider (LKR 1M to 10M) and trigger the Lagrangian solver in real time.
- **Outlet Drilldown Search:** Enables searching specific Outlet IDs to view metadata, cooler count, and spatial gauges.

6.1 Hybrid Explainable AI (XAI) Module

The application integrates an LLM-based component (`src/gold/llm_xai.py`) to generate clear business explanations for outlet scores. The module uses a hybrid architecture:

- **Gemini API Integration:** If `GEMINI_API_KEY` is present, it formats a detailed prompt describing the outlet's potential, growth gap, dynamic tier, cooler count, and spatial gravity signals, requesting a 3-sentence executive summary.
- **Local Fallback Engine:** If offline or no key is present, a deterministic rule-based template engine generates a highly-contextual narrative to ensure zero downtime.

7 GenAI Transparency Log

Team InsightAI utilized Generative AI strictly as a senior thought partner to bridge the gap between abstract mathematics and real-world FMCG supply chain realities.

Table 2: InsightAI GenAI Usage and Prompt Transparency

Workflow Stage	**Human-AI Collaboration Prompts**
1. Business Logic	**Human Prompt:** "If a small shop has potential demand but hasn't got a cooler, we must give him a cooler no... and some shops may need discounts to attract people. We need to correct the logic with real world scenarios applicable not just random math." **AI Impact:** AI transitioned the abstract Knapsack problem into discrete, real-world Trade Packages (90K Core Asset Injection, 15K Discounts).
2. Supply Chain Data Verification	**Human Prompt:** "Why total 5 million doesn't use and no cooler outlets here... still doesn't show 5 million why, pipeline issue?" **AI Impact:** Iterative refinement of the geographic filtering execution sequence to prevent historical distributor migration leakage. The AI corrected the chronological sequence to ensure spatial filtering occurred on the terminal state of the outlet, preventing "ghost allocations" to relocated shops.
3. Explainable AI (XAI)	**Human Prompt:** "Translate these retail metrics [potential, history, coolers, spatial gravity] into 3 business sentences explaining why this shop got its score and recommend action." **AI Impact:** Generated the system prompt for the 'llm_xai.py' dashboard engine.

8 Conclusion and Impact Summary

Team InsightAI has successfully delivered a production-ready spatial analytics and budget optimization pipeline. By resolving the N-Squared spatial indexing bottleneck with SciPy cKDTree and modeling competitive saturation with Huff gravity, we extracted a reliable latent opportunity signal. Quantile regression ($p90$) enabled demand unconstraining by modeling peak performance. The MILP Knapsack optimizer guarantees that the promotional budget is mathematically allocated to maximize incremental volume while obeying strict corporate geographic limits. The Streamlit dashboard bridges these algorithms with intuitive interfaces and LLM explanations, establishing a robust framework for enterprise trade-spend decision-making.